

Leaf-alias: real-time importance sampling of dynamic emissive textures

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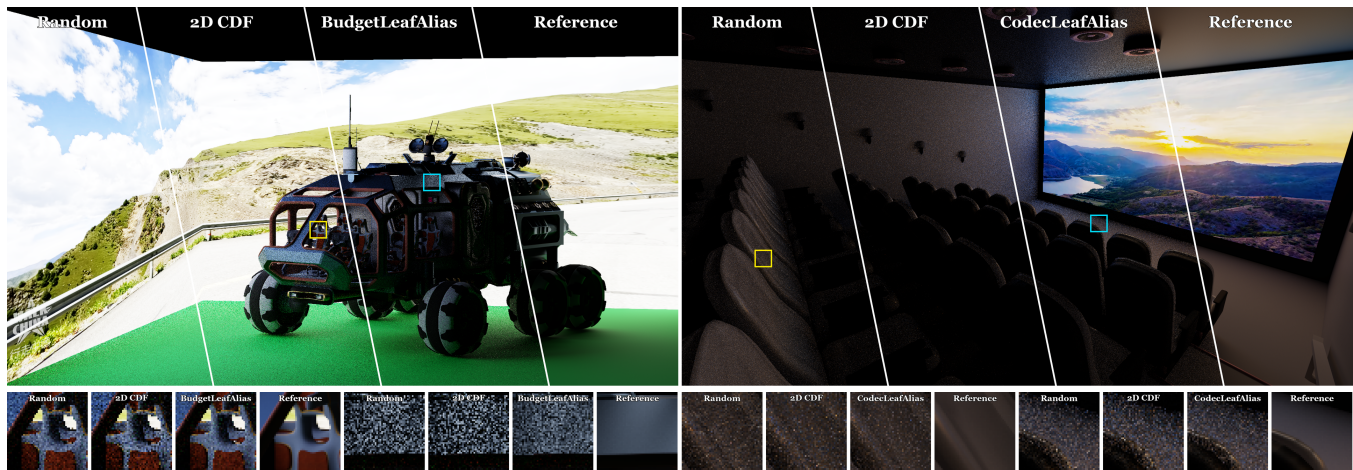


Figure 1: Two scenes lit by a single dynamic, video-driven emissive texture, rendered with our leaf-alias importance sampler. (Left) The Rover scene — a virtual-production LED wall displaying a looping road/mountain video panorama that lights the vehicle directly, sampled here with BudgetLeafAlias. (Right) The Cinema scene — a theater lit only by the animated sunset footage playing on its own screen, sampled with CodecLeafAlias. Each panel compares, left to right: Random sampling, 2D CDF inversion, our technique, and a converged reference, all at their calibrated real-time budgets (≈ 33 ms).

Abstract

We present a method for fast importance sampling HDR textured lights that allows real-time video-driven lighting. Our method partitions a light’s texture into a budgeted, variance driven quadtree and builds an alias table over the resulting leaves, avoiding the cost of a dense per-frame CDF while still capturing the texture’s overall energy distribution. We present two variants: one that reuses the hierarchical structures already present in the metadata of modern codec video streams, and another that builds a budgeted GPU-friendly quadtree to fit the entire sampling structure in L1 cache. Compared to the dense 2D CDF-inversion baseline, at a matched real-time frame-time budget our implemented technique reduces RMSE by 9–53% depending on scene, letting the saved sampling cost be reinvested in additional light samples. It also reduces memory

consumption and transfers by several orders of magnitude when compared to state of the art samplers.

CCS Concepts

• Computing methodologies → Reflectance modeling.

Keywords

These are the keywords

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1 Introduction

Importance sampling of emissive textures is essential for efficient image-based lighting, and increasingly critical in production workflows where video is used directly as a dynamic light source in virtual production and VFX. A single emissive texture lets a renderer reproduce the directional detail of a real environment, far beyond what analytic lights can approximate. Traditionally this texture has been static, captured once and reused for an entire sequence. Increasingly, though, it is video: LED-wall virtual production, AR/VR object insertion, and video-based environment lighting

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all require illumination that changes every frame, not a single fixed capture.

Traditional texture light sampling techniques are designed for static images. They rely on the offline building of large sampling structures that are reused for rendering each frame as long as the image doesn't change. The most common example of such strategy is how rendering and ray tracing toolboxes deal with HDR environment maps. The state-of-the-art approach is to build a two-dimensional piecewise constant distribution from the texture's luminance: a marginal CDF over rows and a per-row conditional CDF. The sampler is then inverted via binary search at sample time, which is expensive for large textures. A more recent and well-adopted approach is to generate a mipmap-style hierarchical importance map [1]. At sample time, a hierarchical descent through a mip pyramid of the texture is performed, recursively choosing one of four children at each level in proportion to their summed weight. This technique is also expensive and requires several iterations per sample, although it is more GPU friendly because the number of iterations is always the same for each sample, unlike in the binary search. Moreover, the mipmap can be fastly constructed using the existing dedicated hardware on modern GPUs. Both techniques require at least $O(N)$ memory for the sampler, with N being the number of pixels.

Another less used but known technique for sampling emissive textures is the Alias method [2]. This technique draws from an arbitrary discrete distribution in $O(1)$ time after an $O(N)$ one-time build, at the cost of losing the CDF's implicit spatial coherence: two adjacent table buckets can alias to two arbitrary, unrelated locations in the original distribution. This fact is known to disrupt the stratification or discrepancy-reduction of the input point set. It also requires $O(N)$ memory to store the correlation between every pixel. The process for building the Alias table, despite being $O(N)$, is difficult to optimize for real-time construction [3].

We present a new sampling method that is fast to build (under 1 millisecond for 4k HDR images), produces a sampler that is tiny, and is $O(1)$ to sample without disrupting the stratification of the Quasi Monte Carlo input. The core idea is to reuse the variance-driven image partition that is already present in modern video codecs, which already compute hierarchical spatial partitions that adapt to local signal variance [Uhrina et al. 2024]. These structures implicitly encode the spatial distribution of luminance and high-frequency content (see Fig. 2). We also propose a variant that builds the partition tree on a budgeted number of leaves, using a Greedy/best-first tree-structured resource allocation [Minasny et al. 2007; Zhou et al. 2024]. Both techniques show consistent quality gains when rendering video light textures in real-time (33 milliseconds per frame), since they allow executing more texture samples without introducing stratification error.

2 Previous works

Importance sampling of emissive textures is a well-studied problem in the graphics literature in the context of image-based lighting. Most of the works focus on pre-processing the images to allow for exact or near-exact sampling without introducing bias [4]. Several methods choose to simplify the textured light into a set of point or area lights [Agarwal et al. 2003], allowing for real-time

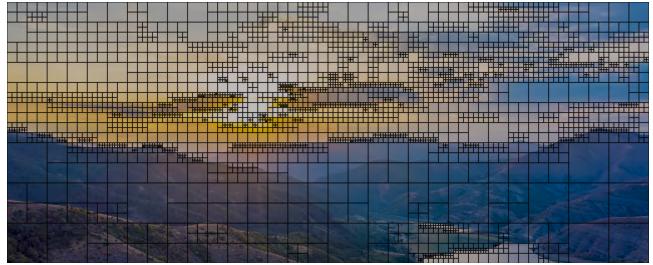


Figure 2: An example of a variance-driven quadtree built by the VP9 video codec. High variance zones are refined into a dense mesh of small cells.

sampling and faster convergence but at the cost of more error in the solution. Cline et al. [Cline et al. 2009] compare several tabular inversion methods, noting the computational order of building each sampler structure and sampling each point. They analyze the 2D CDF inversion via binary search and the per-pixel Alias method, where they flag the disruption of the stratification of the input point set. Per-pixel Alias's raster-flattened table produces a real, measurable periodic banding pattern (Figure 3) that becomes clear in the sample set's power spectrum. We show that our LeafAlias method overcomes those issues because the partition structure helps with stratification.

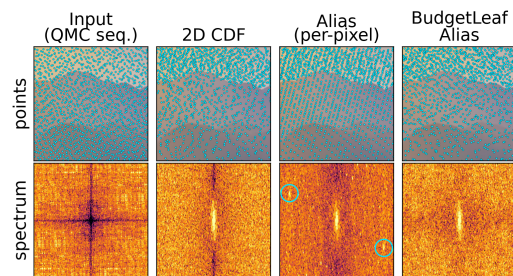


Figure 3: 2D power spectrum of samples from 2D CDF inversion, plain per-pixel alias, and our LeafAlias, from the same Halton input. Per-pixel alias shows a sharp, isolated off-axis peak (flatness 0.270, peak/median 16,113) – a periodic banding artifact from its flat array index resonating with the input sequence's base-2 digit structure. Both CDF inversion and ours stay close to the input's own clean spectrum.

Despite being $O(1)$ to sample, the Alias method is heavy in memory consumption and in the construction of the tables via Vose's technique. Some solutions have been proposed to address the computational performance of those stages [5].

Real-time light sampling has effectively split into two problems: choosing *which* light to sample among many, and choosing *where* to sample within one light's own spatial extent. The first is now mature. Moreau et al. [Moreau et al. 2019] maintain a light-sampling BVH incrementally across frames for thousands of moving lights in under 1ms. Bitterli et al.'s ReSTIR [Bitterli et al. 2020] resamples

across spatial and temporal neighbors to pick among many dynamic lights at a fraction of the cost of resampling from scratch. Conty Estevez and Kulla [Conty Estevez and Kulla 2018] adaptively split a light tree per shading point rather than committing to one fixed cut. All three are real-time, tree- or reservoir-structured, and explicitly designed for scenes that change every frame.

Mesh lights – geometry with thousands of emissive triangles – sit inside this same WHICH-light problem: Moreau et al. and Conty Estevez and Kulla both admit individual triangles as leaf emitters in their trees, each carrying a single flux value. At that scale (10^3 – 10^5 triangles), a plain alias table over per-triangle flux is common practice and cheap to rebuild every frame – e.g. Falcor’s own stock EmissivePowerSampler builds exactly this, one alias-table entry per emissive triangle, rebuilt whenever the light collection changes. This is the same alias method Cline et al. warn about, but at a scale where an $O(N)$ rebuild and $O(N)$ memory are both trivial and the stratification loss is comparatively harmless – picking one of a few thousand triangles, then sampling uniformly inside it, has nowhere near as much structure to disrupt as picking one of millions of texels inside a single texture does. The gap this paper addresses only appears three to four orders of magnitude further out, once “how many things could be picked” stops being thousands and becomes the pixel count of a texture.

The second problem – sampling *where* on a single emissive texture, in proportion to its own spatially-varying radiance – has no equivalent real-time, dynamic-content solution. The standard approach, a 2D piecewise-constant CDF inverted by binary search, is exact but was designed for a texture built once and reused: Agarwal et al. [Agarwal et al. 2003] and Clarberg et al. [Clarberg et al. 2005] both improve on its cost or its ability to sample a *product* of functions, but both still target an expensive, offline-amortized build (Agarwal’s structured point sets are explicitly a preprocess; Clarberg’s wavelet/mip warp is cheaper per-sample but the wavelet tree itself is still built once and reused). Conty Estevez and Lecocq [Conty Estevez and Lecocq 2018] speed up product sampling for production path tracing under the same assumption. For indoor scenes where the environment map is only ever seen through a constrained opening, Bitterli et al. [Bitterli et al. 2015] reparameterize the map so the portal’s visible region becomes an axis-aligned rectangle, sampling the product of radiance and portal visibility directly – again a build meant to be amortized over many static frames, not rebuilt each one. Cline et al. [Cline et al. 2009] instead attack the cost of the lookup itself, comparing binary search against $O(1)$ alternatives including the alias method – but flag a critical cost: “the alias method has a critical flaw that makes it unsuitable for many applications – it tends to disrupt the stratification or discrepancy-reduction efforts that went into creating the input point set.” Every one of these techniques, fast or slow, assumes the distribution does not change from frame to frame.

That assumption is exactly what breaks for video-driven lighting. Havran et al. [Havran et al. 2005] and Iorns and Rhee [Iorns and Rhee 2015] both light scenes from captured video environment maps, but neither targets real-time per-frame resampling of the video itself as the importance distribution. Lu et al. [Lu et al. 2013] do: they rebuild a tabulated per-cube-face CDF every single frame for real-time sampling of a dynamic environment map, using MIS and a per-face sample-count heuristic to cut waste. It is the closest

prior work to our own setting, and it is also exactly where the two halves of this section meet: real-time, yes, but by paying a full CDF rebuild every frame rather than a bounded, cache-fitting structure – the WHICH-light literature’s central idea has not yet reached the WHERE-on-this-light problem. Attempts to make the per-frame rebuild itself cheaper on a GPU point the same direction without closing the gap: Morrical and Zellmann [Morrical and Zellmann 2021] accelerate 2D CDF inversion with ray-tracing hardware at high construction cost and non-trivial implementation complexity, and Lehmann et al. [Lehmann et al. 2021] study GPU-specific weighted sampling and alias-table construction directly.

Two proposals point at the structure we actually build. Ergun et al. [Ergun et al. 2012] importance-sample environment maps with a kd-tree instead of a dense CDF. Lambers [Lambers 2022] proposes a fixed-grid, parameterization-independent sampler and suggests, as future work, replacing the grid with “a quadtree hierarchy... with a split criterion based on the variance of importance values inside a bin” – direct inspiration for the variance-driven quadtree at the core of our technique. Neither is built for real-time, GPU execution, or a texture that changes every frame; our contribution is to take that idea and the WHICH-light literature’s budget-bounded, real-time mindset together, and apply both to a video-rate emissive texture.

Building a tree raises the question Cline et al. posed for the alias method directly: does clustering samples into leaves reintroduce the same stratification loss? We measured this. At the resolutions and leaf budgets the technique targets, leaf-clustering is *not* better stratified than plain per-pixel alias – nearest-neighbor clumping is slightly higher for ours, consistently. What it does avoid is the specific defect Cline et al. warn about: Our technique’s real advantage over both baselines is being far cheaper per sample – an $O(1)$ lookup into a tiny, budget-bounded structure – not better sample placement.

Why CDF cost tracks content, not resolution. GPU cost for tabular CDF inversion – the cost every dynamic-lighting technique above either pays every frame or works to avoid – is driven by luminance *spread*, not pixel count. The marginal CDF is tiny and stays cache-resident regardless of resolution. The conditional CDF is built at full native resolution, but when luminance is concentrated in a few bright regions, most threads’ row lookups land on the same few rows and stay cache-coherent even though the table is large. When luminance is broadly and evenly spread – a photographic backdrop with no dominant hot spot – lookups scatter almost uniformly across a large buffer, thrashing cache regardless of resolution, and the problem worsens as resolution grows. A fixed-size, budget-bounded structure sidesteps this entirely by construction, independent of how the source content is distributed.

Where today’s renderers actually stand. We checked the current open-source implementation of four widely-used renderers directly, rather than assuming from their papers. Falcor [NVIDIA 2024], the framework this work builds on, ships an EnvMapSampler that builds a hierarchical mip-pyramid importance map over the environment texture’s luminance – the Clarberg-style family – built once when the environment map is loaded, with no rebuild path for a texture that changes; and, separately, the triangle-scale alias table (EmissivePowerSampler) described above for mesh lights, which does rebuild whenever the light collection changes, but is sized for

thousands of triangles, not millions of texels. Falcor is also portal-aware without being portal-sampling: its PBRT scene importer parses the portal parameter PBRT files use for Bitterli et al.'s [Bitterli et al. 2015] technique, but explicitly logs it as an unsupported parameter and drops it rather than implementing the reparameterization – so despite recognizing the concept at the file-format level, Falcor does not actually perform portal-masked sampling. **Mitsuba 3** [Nimier-David et al. 2019] builds a Hierarchical2D warp over the environment map – the same mip-pyramid family as Falcor's. **pbrrt-v4** [Pharr et al. 2023] uses a dense PiecewiseConstant2D distribution (the textbook marginal-plus-conditional CDF, inverted by binary search), optionally mean-subtracted (“compensated”) when combined with BSDF sampling via MIS. **Blender Cycles** [Blender Foundation 2024] builds the same kind of dense marginal/conditional CDF over a downsampled background map (`background_map_sample()`). Every one of the four either assumes the environment texture is built once and reused, or pays the full cost of a dense CDF or mip pyramid – none targets a fixed memory/cache budget, and none is built with a texture that changes every frame in mind. This is exactly the gap the rest of this paper addresses.

3 Method

This is the method

3.1 Codec-leaf-alias

3.2 Budget-leaf-alias

4 Results

These are the results

5 Conclusions and future work

These are the conclusions

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